

CON EDISON FLOOR 07 • MUSTERCOMMAND

Life-Safety Operating System • Spatial Headcount, Real-Time Ingress & Cryptographic Evacuation Engine

100% LIFE-SAFETY READINESS

NYC FIRE CODE 3 RCNY §401-06

4 Irving Place, New York, NY • Floor 07

Executive Directive: Elimination of archaic paper clipboards via real-time smartphone QR ingress, spatial CAD denominator tracking, and an immutable SHA-256 ledger.

Target:
195
Occupants

Ingress:
< 10s

Ledger:
SHA-256
Chain

THE 5-STEP LIFE-SAFETY OPERATIONAL LIFECYCLE

STEP 01

Scan & Ingress

Universal smartphone QR scanning over 5G Cloudflare Tunnel or local Wi-Fi. Optical turnstile badge scanning with live SHA-256 ingress stream.

STEP 02

Live Baseline

Spatial headcount denominator across NW, NE, SW, and SE sectors. Dual view: interactive CAD floor roster vs tamper-evident cryptographic block ledger.

STEP 03

Alarm Declare

FSD hazard declaration (Fire, Gas Leak, Smoke, Drill). Instant floor lockdown state machine and emergency multi-channel acoustic siren broadcast.

STEP 04

Push & SitReps

Directives pushed to phones, SMS, PA chimes & displays. 1-tap SOS, Evacuation Chair request, and Warden live Walkie-Talkie (PTT) radio dispatch.

STEP 05

All-Safe Audit

100% headcount closure (195/195). Cryptographic SHA-256 block ledger sealing, FSD Commander signature, and official FDNY 3 RCNY §401-06 Certificate.

How & What AI Operates (Google Gemini 3.6 Flash)

1. Hash-Grounded After-Action Narrative Generator (/api/ai/drill-narrative)

Synthesizes raw snapshot metrics and cryptographic ledger events into official compliance narratives. References specific ledger blocks (e.g., L-0001, L-0028) and benchmarks p95 muster times with zero hallucinations. Built with automatic high-demand fallback.

2. Natural Language Red-List Query Engine (/api/ai/redlist-query)

Allows commanders to ask plain English questions ("Show missing visitors in Sector NW", "Who needs evac chairs?") and instantly converts them into deterministic JSON filter specifications and roster highlights.

3. Natural High-Clarity Public Address (PA) Audio Engine

Replaces robotic synthesis with deliberate dispatcher cadence (0.94 rate, 0.98 natural pitch), commercial harmonic PA chime (E5 → A5), and automatic phonetic emergency expansion (e.g., "FSD" → "Fire Safety Director", "Stair A" → "Staircase Alpha").

Warden Walkie-Talkie (PTT) Distress Radio

• Role-Restricted Emergency Transmission

Strictly gated to verified Floor Wardens (PIN: 2026) and FSD Chief Commanders (PIN: 7007). Unauthorized occupant devices are blocked (403 Forbidden).

• Push-to-Talk (PTT) with Acoustic Effects

Real-time dynamic LED VU meter, key-up radio chirp, roger beep, and squelch tail. Hands-free latch mode for complex hazard briefings.

• Live ADA Distress Transcription & Push

Live Speech-to-Text transcription broadcasts spoken distress words to all 195 occupant devices with real voice playback and permanent ledger logging.

ENTERPRISE TECHNOLOGY STACK MATRIX

LAYER	CORE TECHNOLOGIES	LIFE-SAFETY ARCHITECTURAL ROLE
Frontend	React 19, TypeScript, Tailwind CSS v4, Motion	Sub-16ms responsive commander deck, mobile occupant portal, and high-contrast accessible UI.
Spatial Engine	HTML5 Canvas 2D, Geofencing, Zone Matrix	4-quadrant spatial coordinates, turnstile checkpoints, and designated Area of Rescue Assistance (ARA).
Backend Bus	Node.js (v22+), Express, Server-Sent Events (SSE)	Real-time unidirectional push bus streaming live sign-ins, alarms, and radio distress dispatches in <100ms.
Edge Ingress	Cloudflare Quick Tunnels, Localhost.run Watchdog	Instant public HTTPS URL for off-network 5G smartphones without corporate firewall or VPN friction.
AI Intelligence	Google Gemini 3.6 Flash, Structured JSON Outputs	Deterministic red-list natural query parser and cryptographically grounded After-Action Report narratives.
Security & Audit	Web Crypto API (SHA-256), IndexedDB Offline Queue	Tamper-evident hash-linked event blockchain, offline-resilient action queue, and PIN role authentication.

CERTIFIED FDNY
COMPLIANT

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FLOOR 07

SHA-256 AUDIT LEDGER • GENERATED 2026-09-10 • MUSTERCOMMAND LIFE-
SAFETY OS